

Generative AI and Agentic AI

with
Python



LLM



AGENTS



DATA



AUTOMATION



AI-Powered Learning Journey



Data



Model



Train



Deploy



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GEN AI & AGENTIC AI WITH PYTHON COURSE CONTENT



Course Introduction

- Introduction-What We will Learn In This Course



Importing, Creating Modules & Package

- Import Modules And Packages
- Standard Libraries Overview



Introduction to Python

- Getting Started With Python
- Python Basics-Syntax
- Variables In Python
- Basics Data Types In Python
- Operators In Python



File Handling

- File Operations With Python
- Working with File Paths



Exception Handling

- Exception Handling With try except else and finally blocks



Python Control Flow

- Conditional Statements (if, elif, else)
- Loops



Data Structures Using Python

- Lists and List Comprehension
- Tuples
- Dictionaries
- Sets



Oops

- Classes & Objects
- Single And Multiple Inheritance
- Polymorphism
- Encapsulation
- Abstraction



Functions in Python

- Getting Started With Functions
- Lambda Function In Python
- Map Function In Python
- Filter Functions In Python



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Machine Learning for Natural Language Processing (NLP)



- ➔ Tokenization
- ➔ Text Pre-processing
 - Stemming
 - Lemmatization
 - Stopwords
- ➔ Text Vectorization
 - Bag Of Words
 - N Grams
 - TF-IDF

- ➔ Word Embeddings
 - Word2Vec
 - CBOW
 - Skip Grams
 - GloVe
- ➔ Parts Of Speech Tagging
- ➔ Named Entity Recognition



Deep Learning for (NLP)



- ➔ Welcome to the module on DL
- ➔ Introduction to DL
- ➔ Understanding Deep Learning



Recurrent Neural Networks

- ➔ RNN Forward Propagation with Time
- ➔ Simple RNN Backward Propagation
- ➔ Problems With RNN
- ➔ End-to-End Deep Learning Projects with Simple RNN



Artificial Neural Networks



- ➔ What is a Neuron
- ➔ Activation Functions
- ➔ Step Function
- ➔ Linear Function
- ➔ Sigmoid Function
- ➔ TanH Function
- ➔ ReLU Function
- ➔ Backpropagation vs Forward Pass

- ➔ Gradient Descent
- ➔ ANN Intuition
- ➔ ANN (Hyper Parameter Optimization)
- ➔ Step By Step Training With ANN
 - Optimizer
 - Loss Functions
 - Finding Optimal Hidden Layers & Hidden Neurons In ANN



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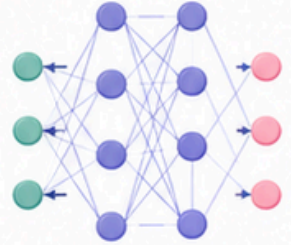


Long Short Term Memory (LSTM)

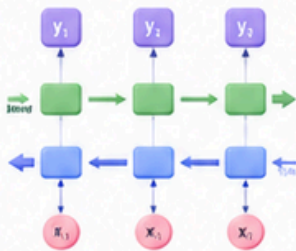


- Why LSTM
- LSTM Architecture
- Forget Gate In LSTM
- Input Gate And Candidate Memory In LSTM
- Output Gate In LSTM

- Training Process In LSTM
- Variants Of LSTM
- GRU RNN Indepth Intuition
- LSTM and GRU End to End Deep
- Learning Project



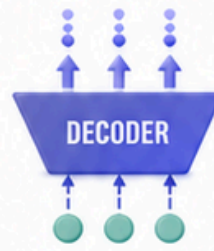
Bidirection RNN



- Bidirectional RNN
- Why To Use It?
- Advantages & disadvantages
- Applications



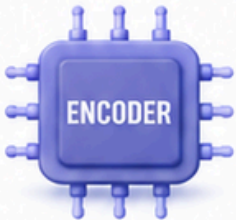
Decoders



- Introduction to Decoders
- Decoder Architecture
- GPT Architecture
- GPT (Masked Multi Head Attention)
- GPT Training



Encoders

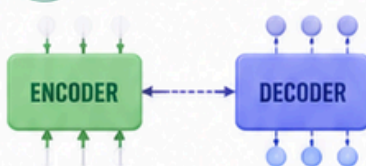


- Introduction to Encoders
- Encoder Architecture
- Introduction to BERT
- BERT Configurations
- BERT Fine Tuning

- BERT Pre Training (Masked LM)
- Input Embeddings BERT
- RoBERTa
- DistilBERT
- AIBERT



Sequence to Sequence Architecture



- Encoder and Decoder
- Indepth Intuition of Encoder & Decoder
- Sequence to Sequence Architecture
- Problems With Encoder and Decoder



Attention Mechanism



- Seq2Seq Networks
- Attention Mechanism Architecture



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Transformers



- ➔ What and Why To Use Transformers
- ➔ Understanding The basic Architecture of Encoder
- ➔ Self Attention Layer Working
- ➔ Multi Head Attention
- ➔ Feed Forward Neural Network With Multi Head Attention
- ➔ Positional Encoding
- ➔ Layer Normalization
- ➔ Layer Normalization Examples
- ➔ Complete Encoder Transformer Architecture
- ➔ Decoder-Plan Of Action
- ➔ Decoder-Masked Multi Head Attention
- ➔ Encoder and Decoder Multi Head Attention
- ➔ Decoder Final Linear And Softmax Layer



Introduction to Gen AI



- ➔ What is Generative AI- AI Vs ML Vs DL Vs Generative AI
- ➔ How Open AI ChatGpt or Llama3 LLM Models are trained
- ➔ Evolution of LLM Models
- ➔ All LLM Models Analysis



Data Preprocessing & Embeddings

- ➔ Data Preprocessing
 - Cleaning
 - embeddings
- ➔ End to end Generative AI Pipeline



Vector Database

- ➔ Vector Databases
- ➔ Vector Index vs Vector Database
- ➔ How Vector db works
- ➔ Vector Database (Practicals)



LLM

Introduction to Large Language Models



- ➔ Introduction to Large Language Models & its architecture
- ➔ In depth intuition of transformer Attention all your need paper
- ➔ How ChatGPT is trained.



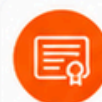
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Complete Guide to OpenAI



- Introduction to OpenAI
- What is OpenAI API and how to generate OpenAI API key?
- Local Environment Setup
- Hands on OpenAI Chat completion API and Completion API

- Function Calling in OpenAI
- Project: Fine-tuning of GPT-3 model for text classification
- Project: Audio Transcript Translation with Whisper
- Project: Image generation with DALL-E



Introduction to Langchain for Generative AI



- Complete Langchain Ecosystem
- Creating Virtual Environment
- Getting Started With Langchain & OpenAI



Open Source LLM

- Introduction to open source LLMs – Llama
- How to use open source LLMs with Langchain
- Custom Website Chatbot using Open source LLMs
- Open Source LLMs - Falcon



Lang Chain – Basic to Advance



- Introduction & Installation setup of langchain
- Prompt Templates in Langchain
- Chains in Langchain
- Langchain Agents and Tools
- Memory in Langchain

- Documents Loader in Langchain
- Multi-Dataframe Agents in Langchain
- How to use Hugging face Open Source LLM with Langchain
- Project: Interview Questions Creator Application
- Project: Custom Website Chatbot



Components & Modules in Langchain



- Introduction To Basic Components And Modules in Langchain
- Data Ingestion With Documents Loaders
- Text Splitting Techniques
 - Recursive Character Text Splitter

- Character Text splitter
- HTML Header Text Splitter
- Recursive Json Splitter
- OpenAI Embedding
 - Ollama Embeddings
 - Huggingface Embeddings



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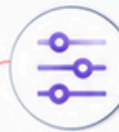


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Retrieval Augmented Generation (RAG)

- ➔ Introduction & Importance of RAG
- ➔ RAG Practical demo
- ➔ RAG Vs Fine-tuning
- ➔ Build a Q&A App with RAG using Gemini Pro and Langchain
- ➔ Retrieval Augmented Generation (RAG)



Fine Tuning LLMs

- ➔ What is fine tuning?
- ➔ Parameter Efficient Fine- Tuning
 - LoRA
 - OLoRA
 - Meta Llama 2 on Custom Data



LlamaIndex - Basic to Advance

- ➔ Introduction to LlamaIndex & end to end Demo
- ➔ Project: Financial Stock Analysis using LlamaIndex



LLM Apps Deployment

- ➔ How to Deploy Generative AI Application
 - Flask
 - AWS



aws



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Introduction to Agentic-AI



- What is Agentic-AI
- AI Agents vs Agentic AI
- Memory & planning



- Agentic Architecture
- Types of Agents
- Mutli Agents



Introduction to LangGraph



- Introduction to LangGraph
- LangGraph Application
- Setting up API Keys
- Developing a simple Graph or Workflow using LangGraph
- LangGraph Components
- Project: End to End Agentic AI Project with LangGraph



Model Context Protocol

- Introduction to Model Context Protocol
- Important components of MCP

- Communication b/w components of MCP
- Building MCP servers with Tools & Client from Scratch using Langchain



n8n



- Introduction to Automation & n8n
- Key features
- n8n Hosting Techniques
- n8n essentials

- n8n nodes
 - Trigger nodes
 - Action nodes
 - Data transformation nodes
 - Switch/filter nodes
 - Split/aggregate nodes



Agentic AI workflow in n8n

- Concept of work flow
- How to give prompts to AI Agent
- Credential setup & testing



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